

**CERTIFICATE OF ANALYSIS**

Natural gas composition and calorific value: 2025 quarterly samples

Report number	SAL-25-GC-0417
Date of issue	3 December 2025
Client	Dunes Energy Company, Southern Dunes Field Central Processing Facility 2 (CPF-2)
Methods	Composition by gas chromatography, ISO 6974-5; calorific value, density and carbon content calculated per ISO 6976:2016
Reference conditions	Combustion 15 °C, metering 15 °C, 101.325 kPa; ideal molar volume 23.645 m ³ /kmol
Expanded uncertainty (k = 2)	NCV ±0.3%; methane ±0.15 mol %; CO ₂ emission factor ±0.5%

Samples

Sample ID	Sampling point	Sampled	Analysed	Sampled by
SAL-25-GC-0417-FG-Q1	SP-3001 fuel gas header	12-Feb-2025	14-Feb-2025	SAL field technician
SAL-25-GC-0417-FL-Q1	SP-5101 HP flare header	12-Feb-2025	14-Feb-2025	SAL field technician
SAL-25-GC-0417-FG-Q2	SP-3001 fuel gas header	14-May-2025	16-May-2025	SAL field technician
SAL-25-GC-0417-FL-Q2	SP-5101 HP flare header	14-May-2025	16-May-2025	SAL field technician
SAL-25-GC-0417-FG-Q3	SP-3001 fuel gas header	13-Aug-2025	15-Aug-2025	SAL field technician
SAL-25-GC-0417-FL-Q3	SP-5101 HP flare header	13-Aug-2025	15-Aug-2025	SAL field technician
SAL-25-GC-0417-FG-Q4	SP-3001 fuel gas header	12-Nov-2025	14-Nov-2025	SAL field technician
SAL-25-GC-0417-FL-Q4	SP-5101 HP flare header	12-Nov-2025	14-Nov-2025	SAL field technician

Results: fuel gas (SP-3001)

Component (mol %)	Q1 12-Feb-2025	Q2 14-May-2025	Q3 13-Aug-2025	Q4 12-Nov-2025	Mean
Methane	88.65	88.97	88.78	89.13	88.88
Ethane	6.49	6.21	6.20	6.12	6.25
Propane	1.53	1.45	1.68	1.50	1.54
i-Butane	0.30	0.31	0.33	0.30	0.31
n-Butane	0.39	0.43	0.37	0.40	0.40
Pentanes plus (as n-C ₅)	0.17	0.21	0.24	0.18	0.20
Carbon dioxide	1.46	1.42	1.41	1.34	1.41
Nitrogen	1.01	1.00	0.99	1.03	1.01
Total	100.00	100.00	100.00	100.00	100.00

Calculated property	Q1 12-Feb-2025	Q2 14-May-2025	Q3 13-Aug-2025	Q4 12-Nov-2025	Mean
Molar mass (kg/kmol)	18.30	18.27	18.33	18.22	18.28
Relative density (air = 1)	0.6318	0.6307	0.6327	0.6289	0.6310
Net calorific value (MJ/Sm ³)	36.35	36.33	36.45	36.28	36.35
Carbon content (wt %)	73.05	73.10	73.16	73.13	73.11
CO ₂ emission factor (t CO ₂ /TJ)	56.99	56.96	57.00	56.89	56.96

Results: HP flare header gas (SP-5101)

Component (mol %)	Q1 12-Feb-2025	Q2 14-May-2025	Q3 13-Aug-2025	Q4 12-Nov-2025	Mean
Methane	76.19	76.04	76.63	76.72	76.39
Ethane	10.93	11.12	10.61	10.45	10.78
Propane	5.87	6.06	6.00	5.90	5.96
i-Butane	1.06	1.11	1.11	1.07	1.09
n-Butane	1.96	1.90	1.94	1.95	1.94
Pentanes plus (as n-C ₅)	0.98	0.96	1.07	1.03	1.01
Carbon dioxide	2.48	2.30	2.18	2.25	2.30
Nitrogen	0.53	0.51	0.46	0.63	0.53
Total	100.00	100.00	100.00	100.00	100.00

Calculated property	Q1 12-Feb-2025	Q2 14-May-2025	Q3 13-Aug-2025	Q4 12-Nov-2025	Mean
Molar mass (kg/kmol)	21.80	21.81	21.76	21.72	21.77
Relative density (air = 1)	0.7527	0.7531	0.7514	0.7498	0.7518
Net calorific value (MJ/Sm ³)	42.29	42.48	42.51	42.27	42.39
Carbon content (wt %)	74.44	74.67	74.83	74.55	74.62
CO ₂ emission factor (t CO ₂ /TJ)	59.47	59.42	59.36	59.35	59.40
CO ₂ per 10 ³ Sm ³ flared at CE 0.98 (t)	2.466	2.474	2.474	2.460	2.468
CH ₄ per 10 ³ Sm ³ vented (t)	0.5169	0.5159	0.5199	0.5205	0.5183

The CO₂ emission factor is calculated as the total carbon (hydrocarbon carbon plus CO₂) multiplied by 44.010 and divided by the net calorific value. Flare factors assume the stated combustion efficiency applied to hydrocarbon carbon; CO₂ present in the gas is emitted unchanged. Results relate only to the samples tested.

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3 December 2025

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